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Article Title: Hyperhydration Induced Decrease in Urinary Luteinizing Hormone Concentrations of Male Athletes may be Compensated by Specific Gravity Adjustment in Doping Control Analysis

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Hyperhydration induced decrease in urinary luteinizing hormone concentrations of male athletes may be compensated by specific gravity adjustment in doping control analysis

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Abstract

Low urinary luteinizing hormone (LH) values have been discussed as a marker to detect steroid abuse. However, suppressed LH concentrations related to highly diluted urine samples could be a misleading indication of anabolic steroid abuse. One aim of the present study was to examine the effect of hyperhydration on the interpretation of LH findings during doping control analysis and to investigate different possibilities to correct volume related changes in urinary LH concentrations. Seven healthy physically active non-smoking Caucasian males were examined for a 72-hour period using water and a commercial sports drink as hyperhydration agents (20 mL/kg body weight). Urine samples were collected and analyzed according to the WADA`s technical documents. Baseline urinary LH concentrations, expressed as mean ± standard deviation, for each individual were within the acceptable physiological range (7.11 ± 5.42 IU/L). Comparison of the measured LH values for both hyperhydration phases (Phase A: 4.24 ± 5.60 IU/L, Phase B: 4.74 ± 4.72 IU/L) with the baseline ('normal') values showed significant differences (Phase A: $P < 0.001$, Phase B: $P < 0.001$) suggesting the clear effect of the urine dilution due to hyperhydration. However, adjustment of urinary LH concentrations by specific gravity based on a reference value of 1.020 seems to adequately correct the hyperhydration induced decrease on LH levels.

Keywords: adjustment, doping control, hyperhydration, luteinizing hormone, specific gravity

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1. Introduction

Luteinizing hormone (LH) is a glycoprotein, with an approximate molecular weight of 28.5 kDa that is physiologically produced and secreted by the gonadotropic cells in the anterior pituitary gland. This heterodimer protein is composed of two polypeptide chains, alpha and beta. Nonetheless, only the heterodimeric α/β has biological activity, which is determined by the hormone-specific β-subunit (Barroso et al., 2012).

In men, LH is responsible for the stimulation of testosterone production by the Leydig cells through binding to and activation of the LH receptors (Handelsman et al., 2009). Circulating LH is subject to negative feedback by the production of endogenous testosterone or the administration of androgens. On the other hand, LH is reported to have negligible effects on blood testosterone levels in women, while it triggers ovulation and induces the synthesis of progesterone and estrogens (Handelsman., 2006, Stenman et al., 2008).

LH can be abused by male athletes in order to increase testosterone production and subsequently enhance physical performance (Stenman et al., 2008). LH administration stimulates the production of endogenous steroids without affecting the ratio of testosterone/epitestosterone, the biomarker that triggers gas chromatography-combustion-isotope ratio mass spectrometry (GC-C-IRMS) analysis for the detection of exogenous anabolic steroids abuse (Goebel et al., 2009). Therefore, LH is included in the World Anti-Doping Agency (WADA) Prohibited List under the class S2 (peptide hormones, growth factors, related substances and mimetics) prohibited both in- and out-of-competition in male athletes (WADA Prohibited List, 2018).

For the determination of LH concentrations in urine, anti-doping laboratories use immunoassays capable of measuring the total content of LH by targeting as many molecular forms as possible. According to the current WADA Technical Document (TD) for reporting and

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management of LH findings in male athletes (WADA TDCG/LH, 2018), any value higher than 60 IU/L (after adjustment if specific gravity (SG) is greater than 1.020) when using the Siemens IMMULITE assay shall be reported as a Presumptive Adverse Analytical Finding (AAF), i.e., a sample test result for which there is a suspicious result without being confirmed. Complementary analysis shall be performed to detect the presence of gonadotropin-releasing hormone (GnRH) analogs, anti-estrogenic substances and aromatase inhibitors in order to report a Presumptive AAF as an AAF, i.e., a confirmed positive result (WADA TDCG/LH, 2018).

Suppressed urinary LH concentrations in male athletes have been discussed for many years as an indication of anabolic steroids abuse, which makes LH a potentially valuable, complementary marker for the interpretation of urinary 'steroid profiles' (Goebel et al., 2009; Handelsman et al., 2014). A threshold had been previously set by WADA (WADA Guidelines CG/LH, 2015) at a concentration of 1.0 IU/L, as a corroborative evidence of LH suppression due to use of exogenous androgens or to indirect androgen doping (Goebel et al., 2009). Values below this threshold had to be considered as an Atypical Finding (ATF), i.e., suppressed LH concentrations with negative results for GnRH, anti-estrogens and aromatase inhibitors interpreted in parallel with the steroid profile markers, compared to normal urine samples collected from the same athlete. Currently, laboratories need to report any case when LH is below the limit of detection (LOD) (WADA TDCG/LH, 2018). During routine anti-doping analysis, it was observed that in many cases urinary LH concentrations close to or below the threshold of 1.0 IU/L were related to low SG values (diluted samples, SG < 1.005).

Based on unreported cases, hyperhydration is being used by a number of athletes as a masking method since urine dilution is not prohibited as per WADA`s current regulations (WADA Prohibited List, 2018). The consumption of high volumes of water or sports drinks leads to

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increased urinary flow rate (UFR) with increased urinary production. Since LH is an endogenous substance, its excretion pattern in urine follows the pattern of urinary production and, therefore, its concentration is influenced by the renal function and the UFR (Stenman et al., 2008). As already mentioned, according to the WADA TD2018CG/LH (WADA TDCG/LH, 2018), adjustment based on SG is applied only to samples with SG greater than 1.020. However, in our opinion, to reduce any dilution effect after hyperhydration on the urinary LH profile, adjustment of LH concentrations related to low SG should also be applied to normalize values close to the baseline urinary LH profile.

In this context, different adjustment methods were applied in this study to reduce the variability induced by the urine dilution due to hyperhydration. Furthermore, in the present study, as part of a WADA research grant (10D21CG), we aimed to assess whether urine samples with low SG caused by the consumption of high volumes of fluids (hyperhydration) can affect the interpretation of LH findings by altering the LH profile, leading to potentially suspicious (false positive) findings. To this end, the threshold of 1.0 IU/L for low urinary LH concentrations (WADA Guidelines CG/LH, 2015) was used to allow statistical comparisons.

2. Experimental

2.1 Study population

Seven healthy, recreationally active, non-smoking Caucasian male participants (n=7) were recruited for the present study with the following physiological parameters (as means ± standard deviation (SD)): 38.4 ± 8.2 years, 88 ± 17 kg body weight (BW), 178 ± 6 cm height, 2.1 ± 0.3 m^2 body surface area and 136 ± 19 mL/min creatinine clearance. A complete medical history was collected by interview with each participant to verify the initial inclusion criteria as part of the clinical trial (Table 1). All the participants underwent a complete medical screening procedure

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including physical and haematological examinations to ensure their healthy state and were found eligible to participate in the present study. Written consent was provided by all the participants prior to their participation and they were free to withdraw from the study at any time. The study was approved by the Ethics Committee of the Anti-Doping Lab Qatar, Doha, Qatar (F2013000006) and performed in accordance with the Declaration of Helsinki.

2.2 Study design

Prior to inclusion and during the study, each participant was asked to maintain his normal training protocol and dietary program, as submitted and approved before the beginning of the study, but avoid anything that might interact with circulatory, gastrointestinal, hepatic or renal function (i.e., alcohol, caffeine-containing beverages, grapefruit juice and diuretic substances).

Participants were instructed to arrive at the testing venue between 7:00 to 8:00 am following at least an eight-hour overnight fast. Upon arrival, a urine sample was collected and hyperhydration was implemented by either water (Phase A) or a commercial sports drink (Phase B) within 30 minutes at a volume of 20 mL per kg BW (i.e., 1400 mL for a 70-kg male). During the study development phase, three hyperhydration levels were examined: low (10 mL/kg BW) (Donike and Rauth, 1992), medium (20 mL/kg BW) and high (30 mL/kg BW). Based on these preliminary (unpublished) data, it was concluded that the high level of hydration (30 mL/kg BW) was impossible to be consumed, due to the discomfort caused. A carbonated balanced glucose-electrolyte solution with the following characteristics: 6.0% w/v carbohydrate, 104.6 kJ, Na 0.05% w/v, K 0.012% w/v, pH 3.6 and an osmolality of 438 mOsm/kg was chosen as the hyperhydration agent of phase B. Both hyperhydration agents were kept at room temperature. After hyperhydration, a 15-minute resting interval was taken and a second urine sample was collected. During the resting interval, participants were in a sitting posture at room temperature, without

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ingesting any sort of fluids. BW, blood pressure, heart rate and respiration rate were measured before and after ingestion and exercise. Hyperhydration was repeated at the testing venue 24 and 48 hours after first ingestion on three consecutive days per hyperhydration agent i.e., a total of 72-hour. Urine sample collection was performed at each voluntary micturition throughout each hyperhydration phase. Both hyperhydration phases were completed by all the participants. A four-day wash-out period was introduced between the phases to avoid possible carry over effects.

Prior to phases A and B (hyperhydration phases), each participant was asked to collect his urine samples for a 24-hour period maintaining his normal dietary routine. These samples were used to establish a baseline ('normal') endogenous LH profile per participant.

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2.3 Sample collection and preliminary measurements

Urine samples were collected into 200 mL sterile containers at each voluntary micturition according to the study protocol (section 2.2). Time of collection and volume of each urine sample was recorded prior to refrigeration and used for the UFR calculation. Preliminary laboratory measurements, including pH, SG, conductivity (mS/cm) and creatinine (CRE, mmoL/L) were performed for all the urine samples to assess hyperhydration influence and possible correlation with measured endogenous LH concentrations.

pH was measured by a 700-pH meter (Orion Star A111, Thermo Fisher Scientific, Waltham, MA, USA) and SG was determined using an Atago 3464 Refractometer (Atago, Tokyo, Japan). Conductivity was determined using an HQ14d Portable Conductivity Meter (Hach Company, Loveland, CO, USA). A Cobas INTEGRA 400 plus system (Roche Diagnostics, Rotkreuz, Switzerland) was used for the measurement of CRE.

After preliminary laboratory measurements, urine samples were stored at -30 °C until the measurement of LH concentrations.

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2.4 Sample analysis

Urinary LH concentrations were quantified by an automated IMMULITE® analyzer (Siemens, Fladers, USA) using a solid-phase, two-site chemiluminescent enzyme immunometric assay according to the specifications of the manufacturer. The pre-analytical procedure was performed according to the WADA TD2018 CG/LH (WADA TDCG/LH, 2018).

As the IMMULITE LH assay was developed for measuring LH concentrations in serum samples, the implementation of LH detection in urine samples for doping control analysis was validated according to the WADA International Standard for Laboratories (WADA ISL, 2016) and the WADA TD2018 CG/LH (WADA TDCG/LH, 2018). Linearity was established by serial dilutions of a high concentration (40.0 IU/L) and a low concentration (2.0 IU/L) urine samples and analysis of replicate measurements at each dilution. Correlation coefficient was above 0.99 with a slope of 1.02. Intermediate precision (expressed as percent coefficient of variation) was below 6.50% evaluated by the analysis of six replicates of three different urine samples on the same day, and on three different days by different analysts and different lots of LH immunoassay kits, respectively. Specificity of the assay was determined by analyzing NIBSC Reference materials to test the method`s cross-reactivity with molecules of closely related structures. Ten different urine samples with measured LH concentrations were spiked with a known LH concentration at a 1:1 ratio and were analyzed again for the evaluation of matrix interferences. No carryover was observed after analysis of a urine sample with high LH concentration followed by a low concentration urine sample.

2.5 Data analysis

Adjustment of urinary LH concentrations to compensate the hyperhydration effect was performed following the equations presented in Figure 1. The 'conventional' SG-adjustment

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method is based on the Levine-Fahy equation (Levine and Fahy, 1945) (Figure 1, Equation 1) and is used by WADA as the official adjustment method. The non-adjusted (measured) concentration (C, IU/L) is adjusted to a population reference SG value of 1.020 resulting in the SG-adjusted concentration (C_{SG}, IU/L). Alternative adjustment methods were selected from the literature (Heavner et al., 2006; Middleton et al., 2016), applied and compared to the 'conventional' SG-adjustment method. Equation 2 (Figure 1) demonstrates a log-linear relationship between the UFR and the non-adjusted concentrations referred to as the UFR-adjustment equation proposed by Araki *et al*. (Araki et al., 1990). The parameter b is the slope of this log-linear relationship referred to as Araki`s b slope and is analyte dependent.

A modified SG-adjustment method ($SG_{(z)}$-adjustment) based on physiological aspects was proposed by Vij and Howell in 1998 (Vij and Howell, 1998) introducing the value z through the Equation 3 (Figure 1). The parameter z is the slope of the log-linear relationship between the non-adjusted concentration and the SG, expressed as SG-1 and it is also analyte dependent. The b, z, values were established for the study population and both phases A and B. Araki`s b values were equal to 0.770 ± 0.234 (Phase A) and 0.795 ± 0.185 (Phase B) and Vij and Howell`s z values were 1.02 ± 0.295 (Phase A) and 0.965 ± 0.161 (Phase B), expressed as mean ± SD.

Adjustment methods based on CRE concentration were also applied (Boeniger et al., 1993). The CRE-adjustment method (Boeniger et al., 1993), is based on the correction of the measured concentrations using a reference CRE value and the CRE concentration per sample. The reference CRE value is the average value of the CRE concentrations calculated from the study population per each phase (Equation 4). Thompson *et al*. proposed the [CRE]-regression-adjustment (Equation 5) based on the log-linear relationship between the non-adjusted concentrations and the CRE concentrations (Thompson et al., 1990). According to the Equation 5, m is the slope of the

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log (C, IU/L) = f (log (CRE, mmoL/L)) linear least squares regression and n is the log value of the average urinary CRE concentration per study phase. Adjusted concentrations are resulted by inversion of the logarithmic function of the Equation 5.

In the context of the present study, adjustment was performed to all the measured LH concentrations obtained during the phases A and B using Equations 1–5 including the baseline values.

2.6 Statistical analysis

Data distribution was tested using the Kolmogorov-Smirnov (K-S) normality test. Significance was set at $P < 0.05$ level and all tests were two-tailed with 95% confidence intervals (CI).

Outlier detection on the individual baseline data occurred applying the interquartile range (IQR) using a step of 1.5 x IQR. Box and whisker plots were also established to investigate possible outliers. Nevertheless, no outliers were removed from the statistics. The K-S test results revealed that the data followed a non-normal distribution even after a logarithmic transformation. The Mann-Whitney U test was used to detect possible significant differences between the baseline and the hyperhydration phases (A and B) and, therefore, to confirm the dilution effect on the LH concentrations due to hyperhydration. Comparisons of the non-adjusted and the adjusted concentrations obtained by each method (Figure 1) were performed by the Wilcoxon signed-rank test to evaluate the performance of the different adjustment methods. Pearson correlation coefficients (r_p) were calculated to determine the strength of the relationship of the obtained LH concentrations against the values of the measured urinary parameters namely, pH, SG, UFR, CRE and conductivity, and used as supportive tool to assess the hyperhydration influence.

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All the comparisons were performed from pooled data of all the participants (referred to as study population data later in this text).

Data analysis was performed by SPSS version 25.0 (IBM SPSS Statistics for Windows, Version 25.0, IBM Corporation, Armonk, NY, USA) software package.

3. Results and discussion

All the described procedures were implemented as referred to in the approved clinical protocol without any kind of deviations. The scheduled volume of each hyperhydration agent was well tolerated by all the participants.

Influence of hyperhydration on LH concentrations

During the 24-hour baseline collection scheme, endogenous LH concentration levels were clearly within the acceptable physiological range (7.11 ± 5.42 IU/L). Urinary LH concentrations followed the rhythm of urinary production expressed by the pattern of the measured urinary parameters (pH, SG, UFR, CRE and conductivity) as presented in Figure 2. Conductivity is considered one of the most widely used hydration markers (Armstrong et al., 2007) and, therefore, it was examined as an extra marker even though it is not applied in routine doping control analysis as pH and SG (WADA Urine Sample Collection Guidelines, 2014). The 'conventional' SG-adjustment method (Figure 1, Equation 1) shall be applied to all the samples with a SG greater than 1.020 as specified by the WADA TD2018 CG/LH. However, to examine the equivalence of the adjustment methods used in the present study (Figure 1, Equations 1–5), all the baseline concentrations were adjusted by all the methods and the obtained adjusted values were statistically compared. No significant difference was observed between any of the adjustment methods since there is no dilution effect on the baseline data. Baseline values adjusted by the 'conventional' SG-

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method for all the samples were used as controls indicating the normal LH profile of the study population (referred to as normal baseline values later in this paper).

Urinary LH concentration-time profiles are illustrated in Figure 3 for the baseline and the hyperhydration phases A (water) and B (sports drink) during the first 24 hours after hyperhydration. Deviations from the normal baseline values were demonstrated and were more intense immediately after excessive fluid intake. The resulting concentration values, expressed as mean ± SD, were 7.72 ± 5.42 IU/L, 4.24 ± 5.60 IU/L and 4.74 ± 4.72 IU/L for the baseline phase and the phases A and B, respectively. The measured LH concentrations returned to levels close to baseline values approximately 10 hours after first fluid intake (Figure 3). The same pattern was observed again after second and third hyperhydration implementation during the 72-hour sample collection scheme. Comparison of the non-adjusted LH values with the normal baseline values showed significant differences ($P < 0.001$ Phases A and B) suggesting a clear effect of the urine dilution due to hyperhydration during the first 24 hours as well as for the whole duration (72 hours) of each phase.

Reduced LH values observed immediately after intake of each of the hyperhydration agents (Figure 3) were related to a higher UFR and lower recorded SG, CRE and conductivity values. More specifically, UFR ranged between 0.25 and 4.76 mL/min (median 1.12 mL/min) at baseline phase (no hyperhydration), between 0.18 and 43.8 mL/min (median 2.50 mL/min) at phase A and between 0.28 and 22.3 mL/min (median 2.44 mL/min) at phase B. The linear relationship between LH concentrations and urinary parameters (pH, UFR, CRE and conductivity) was determined and expressed by Pearson`s correlation coefficients. The measured LH concentration values were positively correlated with both SG values (r_p: 0.37–0.51, $P < 0.001$, 95% CI: 0.25–0.60) and CRE values (r_p: 0.62–0.66, $P < 0.001$, 95% CI: 0.53–0.72) revealing that hyperhydration influences in

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the same manner these two urinary parameters and the measured LH concentrations. As expected, a negative correlation was recorded between LH and UFR (r_p: -0.49 to -0.35, $P < 0.001$, 95% CI: -0.46 to -0.22), since excessive fluid intake leads to higher urine volumes in shorter time intervals and subsequently to higher UFR values and to urine samples with lower LH concentrations. Weak correlations with r_p values ranging from -0.009 to -0.035 ($P < 0.001$, 95% CI: -0.12–0.17) were observed between LH values and pH in all the study phases which was expected due to the urine buffer capacity.

In the current WADA TD2018 CG/LH (WADA TDCG/LH, 2018), any case of LH below LOD has to be reported since it is not applicable to the threshold of 1.0 IU/L (WADA Guidelines CG/LH, 2015). Detection of LH at concentration levels below 1.0 IU/L may serve as an indication of LH suppression due to use of exogenous androgens or to indirect androgen doping (Goebel et al., 2009). Therefore, any sample below that threshold should be considered as an ATF compared to normal urine samples collected from the same athlete. In addition, these values should be interpreted in parallel with the values obtained for the urinary steroid profile.

However, LH values below the threshold of 1.0 IU/L could also be related to highly diluted samples (SG < 1.005) resulting in potentially suspicious (false positives) findings. Increased urine volumes were recorded immediately after hyperhydration resulting in an altered LH profile compared to the "normal" lower LH concentrations (IU/L) with many cases below the threshold (Figure 3). Nevertheless, no significant difference (Phase A: $P = 0.87$, Phase B: $P = 0.84$) was observed on the urinary 24-hour LH cumulative amount between the baseline and the hyperhydration phases revealing that excessive fluid intake has no effect on the urinary LH profile (Figure 4). The same pattern was also observed during the 48 and 72 hours sample collection. Therefore, both hyperhydration agents led to decreased measures of urinary LH concentrations at

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early points after hyperhydration due to the increased urine volumes resulting in LH values below the threshold of 1.0 IU/L. Water intake had a more intense and lasting effect compared to the sports drink probably due to the higher energy content of the later, resulting in delayed gastric emptying and lower absorption at early times after hyperhydration (Okabe et al., 2015). During the baseline phase, all the samples were reported with LH values above the threshold of 1.0 IU/L while 24.6% and 15.6% of the measured LH values were below 1.0 IU/L during the first 24 hours of the hyperhydration phases A and B, respectively.

Therefore, to minimize the observed dilution effect due to hyperhydration on the LH concentration values, adjustment is of utmost importance.

Different adjustment methods were applied in order to estimate if any other parameter except for the SG could compensate the dilution effect by adjusting urinary LH concentrations close to the baseline urinary LH profile. Descriptive statistics are reported in Table 2, including the minimum, maximum, median, mean and percentiles (25 and 75%) of the resulting adjusted concentration values by the different adjustment methods (Equations 1–5).

Comparison of the non-adjusted concentrations with the SG, UFR and $SG_{(z)}$-adjusted concentration values showed significant differences ($P < 0.001$). On the other hand, the adjustment methods based on CRE (Equations 4 and 5) did not present any significant difference (P range 0.23–0.92) compared to the non-adjusted values and, therefore, no compensation on the dilution effect.

Furthermore, the calculated percentages (%) of samples with LH concentrations below the threshold of 1.0 IU/L are shown in Table 3 per adjustment method for both phases A and B. SG and UFR gave the lower % of samples for the water phase (Phase A) while SG and $SG_{(z)}$ adjustment methods showed the same percentage of 0.53% during the sports drink phase (Phase B). On the

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other hand, the CRE-based adjustment methods (Equations 4 and 5) present the highest % of samples below the threshold compared to the other methods. Combining the results of both phases, the SG-adjustment method seems to act more efficiently on the correction of the dilution effect. Furthermore, the UFR-adjustment method, in terms of practical application, is more time-consuming requiring at least two consecutive urine samples for the calculation of the UFR and then the adjusted concentrations. On the other hand, the adjustment methods based on CRE require extra equipment that it is not included in the routine parameters measured in the collected samples like SG and pH. Finally, according to Vij and Howell (Vij and Howell, 1998) when the z values are equal or close to 1 (range 0.855–1.10), the $SG_{(z)}$ and SG methods are equivalent. Therefore, taking into account the ability to apply the adjustment method in routine practice (time, cost, etc.) and the results obtained within this study, the SG-method applied by WADA compensates efficiently for the hyperhydration effect.

As reported recently by Athanasiadou *et al*., the SG-adjustment method was also found to compensate any effect caused by dilution on the urinary endogenous steroid profile (Athanasiadou et al., 2018). Results showed that excessive fluid intake can alter the urinary steroid profile markers, with the steroid ratios, however, remaining unaffected. Similarly, in the study of Haase *et al.* it is reported that SG-adjustment of urinary concentrations of salbutamol after inhaled administration of the maximum dose permitted by WADA, was found useful for samples with SG below 1.020 g/mL, but less useful when SG exceeded 1.020 g/mL (Haase *et al.*, 2016).

The performance of the SG-adjustment method was also evaluated by comparing the normal baseline values with the SG-adjusted values for both phases A and B. No significant difference (Phase A: $P = 0.08$, Phase B: $P = 0.86$) was observed between these values indicating the adequacy of the selected method to adjust the values close to the baseline urinary LH profile.

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Despite our promising results, there are several limitations in the present study. First of all, a larger number of participants would probably reduce the observed variability that could probably mask possible differences between the administration phases. In addition, the use of controlled/standardized diet during the study would probably further decrease the observed variability. However, and despite these limitations, the results of the study are quite clear and promising regarding the effect of hyperhydration on urinary LH levels. Furthermore, it was clearly showed the superiority of the official WADA SG-adjustment method compared to the UFR and CRE based methods for the compensation of the dilution effect due to hyperhydration.

In this study, we focused on the effect of hyperhydration on the urinary endogenous LH levels without any intervention that would lead to either elevated > 60 IU/L (e.g., GnRH) or below the threshold (1.0 IU/L) LH levels (e.g., androgen administration). Our goal was to examine if hyperhydration could influence the excretion of endogenous LH by leading to concentrations below the threshold that could be attributed as false positive findings. However, work is ongoing to explore the results of the present study and investigate their applicability for doping control purposes. To this end, the hyperhydration effect on the detection sensitivity and pharmacokinetic parameters of representative prohibited substances is considered.

4. Conclusion

Excessive fluid consumption alters the urinary LH profile leading to low concentrations immediately after intake of different hyperhydration agents. Therefore, suppressed LH concentrations could lead to potentially suspicious (false positive) findings, when being interpreted as related to androgen abuse. Hyperhydration induced decrease in urinary LH concentration levels was found to be corrected near the baseline values by adjusting the measured concentrations. The 'conventional' SG-adjustment method appeared to be most effective in terms of practical

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application and management of the hyperhydration induced dilution effect by adjusting the values close to the baseline urinary LH profile. Based on the results of the present study, SG-adjustment of urinary LH concentrations independent of the initial SG of the urine sample could be useful.

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Conflict of interest

The authors have no conflict of interest to declare.

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A. 'Conventional' adjustment method

$$C_{SG}\left(\frac{IU}{L}\right) = C\left(\frac{IU}{L}\right) * \left(\frac{1.020 - 1}{SG_{sample} - 1}\right) \quad (1)$$

B. UFR-adjustment method

$$C_b\left(\frac{IU}{L}\right) = C\left(\frac{IU}{L}\right) * UFR\left(\frac{mL}{min}\right)^b \quad (2)$$

C. SG(z)-adjustment method

$$C_{SG(z)}\left(\frac{IU}{L}\right) = C\left(\frac{IU}{L}\right) * \left(\frac{1.020 - 1}{SG_{sample} - 1}\right)^z \quad (3)$$

D. CRE-adjustment method

$$C_{CRE}\left(\frac{IU}{L}\right) = C\left(\frac{IU}{L}\right) * \left(\frac{CRE_{ref}\left(\frac{mmoL}{L}\right)}{CRE_{sample}\left(\frac{mmoL}{L}\right)}\right) \quad (4)$$

E. [CRE]-regression-adjustment method

$$Log\left(C\left(\frac{IU}{L}\right)\right) CRE_{regression} = Log\left(C\left(\frac{IU}{L}\right)\right) + \left(m * \left(Log\left(CRE_{sample}\left(\frac{mmoL}{L}\right) - n\left(\frac{mmoL}{L}\right)\right)\right)\right) \quad (5)$$

Figure 1.

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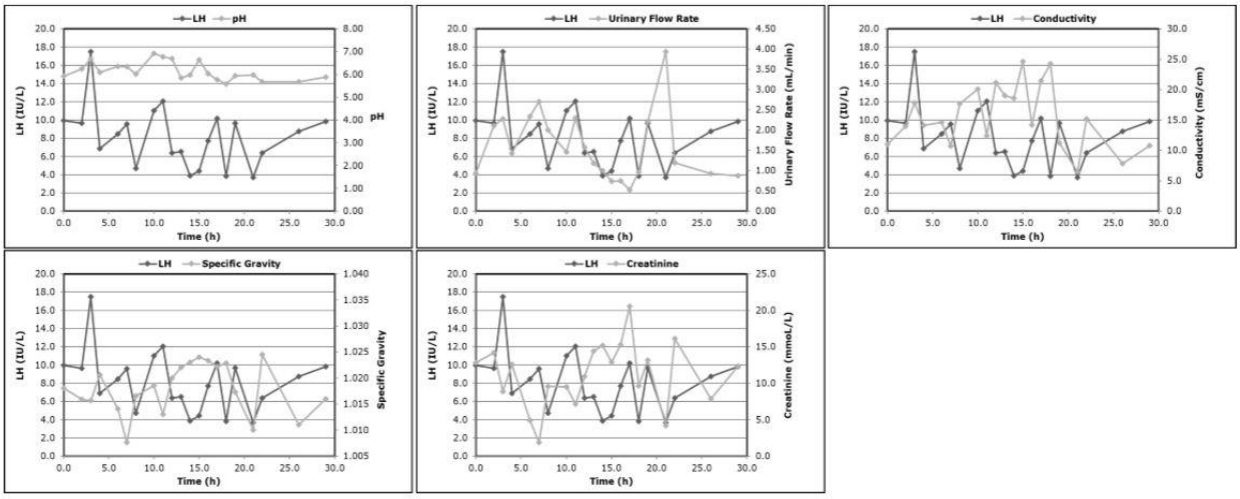

Figure 2.

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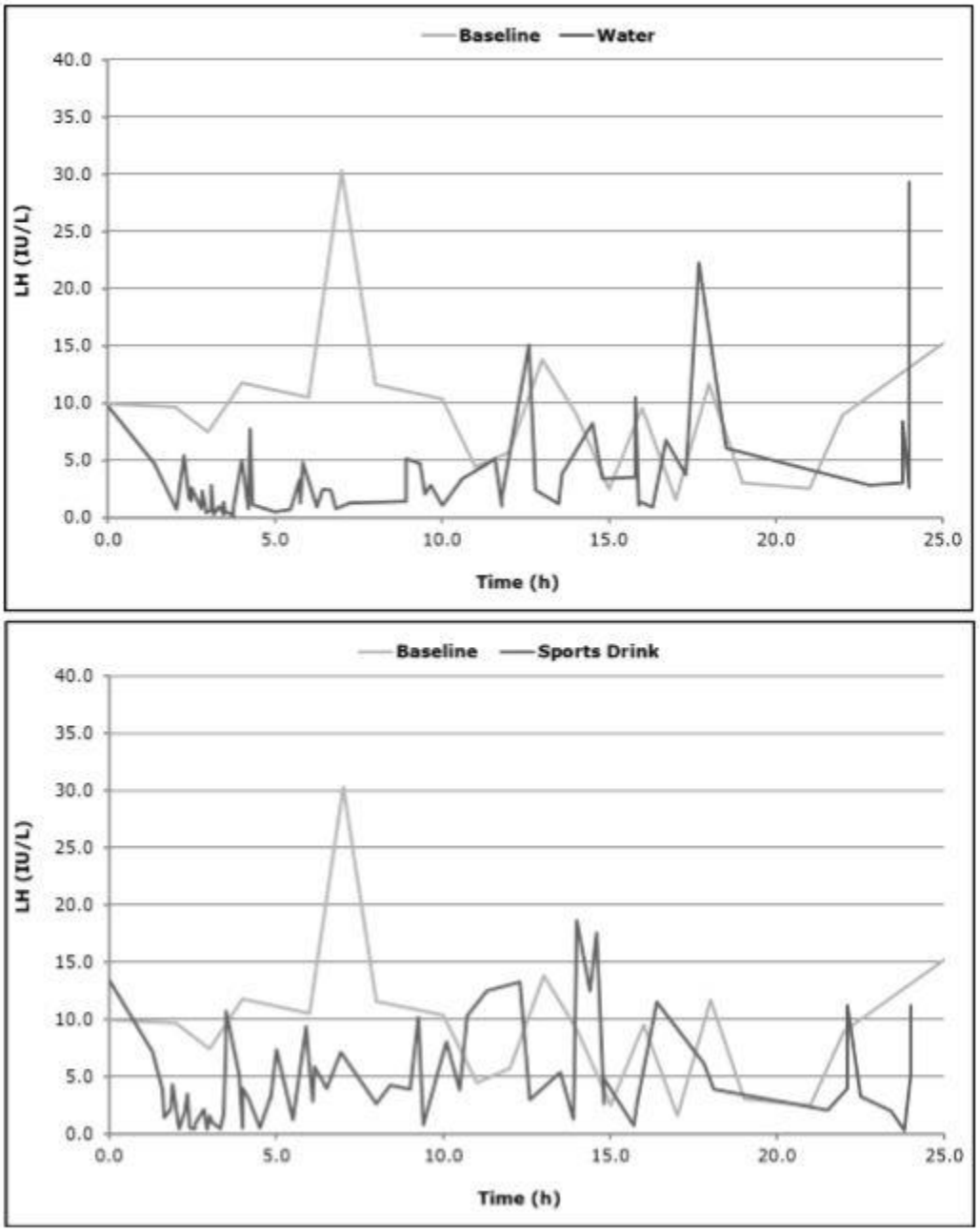

Figure 3.

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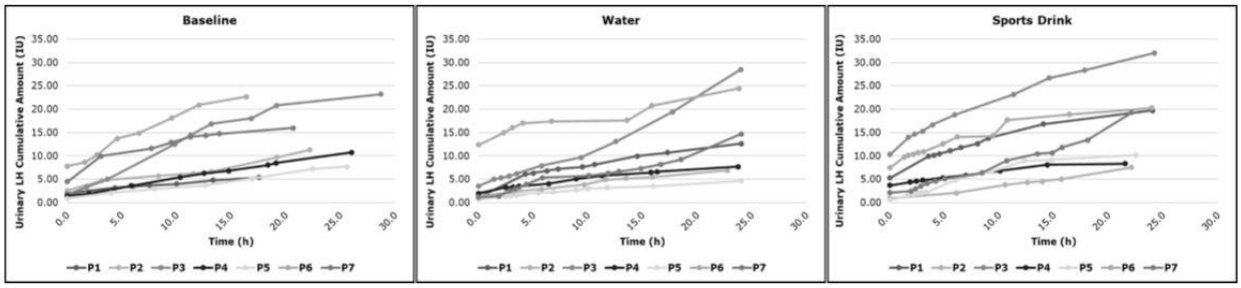

Figure 4.

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Table 1. Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
healthy	History of hypersensitivity to the examined drug(s) or allergy;
recreational athletes	Any clinically significant acute or chronic disease that can affect the absorption, metabolism, distribution and excretion of the test drug(s);
between 20 and 55 years;	Acute or chronic abuse of alcohol;
Non-smokers	Drug abuse or excessive consumption of drugs (amphetamines, opiates, benzodiazepines, cannabinoids, barbiturates, cocaine and tricyclic anti-depressive agents)
	Hyponatraemia and urinary tract infection;
	Known carrier of HBs antigen, HBc antibodies or anti-HIV antibodies, hepatitis C;
	Positive alcohol breath test (20 mg/100 mL of blood);
	Positive drug urinary screening analysis;
	Any pharmaceutical treatment including OTCs, vitamins and/or supplements (except paracetamol for pain relief) two weeks prior to inclusion and during the study;

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Table 2. Descriptive statistics of urinary luteinizing hormone (IU/L) obtained from all the adjustment methods per study phase for the total duration of 72 hours.

Phase A: Water						
Adjustment Method	**Mean**	**Median**	**Minimum**	**Maximum**	**Percentiles 25%**	**Percentiles 75%**
Non-adjusted*	4.24	2.44	0.120	41.5	1.06	4.93
SG-adjusted	7.87	5.29	0.800	41.6	2.82	11.1
UFR-adjusted	7.43	5.07	0.980	54.4	3.10	9.16
SG(z)-adjusted	6.53	4.24	0.520	39.5	2.38	8.42
CRE-adjusted	4.91	3.64	0.480	19.2	1.97	6.32
CRE-regression adjusted	4.40	3.24	0.420	21.5	1.86	6.33
Phase B: Sports Drink						
Adjustment Method	**Mean**	**Median**	**Minimum**	**Maximum**	**Percentiles 25%**	**Percentiles 75%**
Non-adjusted*	4.74	3.24	0.200	34.4	1.36	6.86
SG-adjusted	8.64	7.64	0.880	36.4	4.03	11.4
UFR-adjusted	8.04	6.66	0.770	31.3	3.41	11.0
SG(z)-adjusted	7.95	6.83	0.920	34.4	3.63	10.6
CRE-adjusted	6.44	5.33	0.560	30.7	3.04	8.96
CRE-regression adjusted	5.52	4.85	0.690	25.9	2.64	7.63

SG: *specific gravity*, UFR: *urinary flow rate*, CRE: *creatinine***in all cases significant differences compared to the adjusted data was observed (P<0.001, 95% CI) excepted for the CRE and CRE-regression adjusted values.*

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Table 3. Percentages of endogenous luteinizing hormone concentrations below the threshold of 1.0 IU/L per adjustment method during the first 24 hours after hyperhydration and for the whole duration (72 hours) of each phase (A: Water, B: Sports Drink).

Hyperhydration Phase	Non-adjusted	SG-adjusted	UFR-adjusted	SG(z)-adjusted	CRE-adjusted	CRE-regression adjusted
Phase A (24 h)	24.6%	3.08%	1.54%	3.08%	10.8%	15.4%
Phase B (24 h)	15.6%	1.56%	3.13%	1.56%	4.69%	3.13%
Phase A (72 h)	22.6%	1.61%	1.08%	2.69%	8.06%	8.06%
Phase B (72 h)	18.9%	0.53%	1.05%	0.53%	3.68%	2.63%

SG: *specific gravity*, UFR: *urinary flow rate*, CRE: *creatinine*

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